

(CART) Model Comparison Project Write Up Summary

Which model performed best, and by how much?

The linear models basically tied for first place. Linear regression and Lasso both got a CV RMSE of 1.00, with Ridge close behind at 1.03. On the test set, Lasso barely edged out linear regression (1.139 vs 1.142) but the difference is so small it's basically a tie.

The surprise here is that Random Forest came in second to last with a CV RMSE of 1.54 and test RMSE of 1.66. I expected it to win since it's supposed to be this powerful ensemble method, but it got destroyed by plain linear regression. CART was the worst at 2.20, which makes sense since it's just one tree with limited flexibility.

So the ranking from best to worst is:

1. Linear Regression / Lasso (basically tied)
2. Ridge
3. SVM
4. Random Forest
5. CART

The best model (linear regression) beats the worst model (CART) by more than 1 full RMSE point, which is a huge difference. But it only beats Random Forest by about 0.5 points.

Were there meaningful differences in stability?

Looking at the standard deviations, the linear models were the most stable across the CV folds. Linear regression and Lasso both had SD of about 0.13, meaning their performance was pretty consistent no matter which fold was held out.

CART was the least stable with SD of 0.32 which is almost 2.5x higher than the linear models. Random Forest was also relatively unstable at 0.23. This makes sense because tree-based models can change a lot depending on which data points they see.

So not only did the linear models have the best average performance, they were also the most reliable. That's a double win.

What tradeoffs do you see between interpretability and performance?

This dataset shows that you don't always have to sacrifice interpretability for performance. The most interpretable models (linear regression, lasso, ridge) were also the best performers. With linear regression, I can look at the coefficients and say exactly how much each variable affects Sales. With Lasso, I can even see which variables got shrunk to zero and removed.

Random Forest is way harder to interpret and it's 500 trees combined together, so you can't really look at it and understand what it's doing. And in this case, all that complexity actually hurt performance instead of helping.

I think the reason the linear models won is that the relationship between the predictors and Sales is mostly linear in this dataset. When the true pattern is linear, adding nonlinear models just introduces extra noise and overfitting. The fancy models are trying to find complex patterns that don't actually exist.

Plots:

